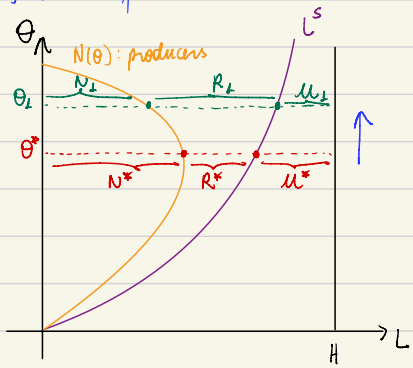
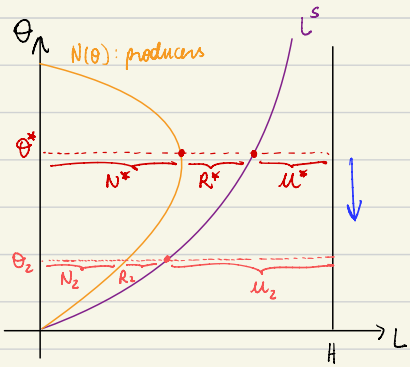
Quiz 6

1) Costs of $\downarrow U$:



In order to attain $u_1 < u^*$, we need to bear $R_1 > R^*$ and $N_1 < N^*$.

2) Costs of $\uparrow U$



When we raise U, both N_2 and R_2 decrease.
$\downarrow$ # employed workers.

3) Efficient unemployment rate: it maximizes the social welfare.
Since the social planner is subject to matching function such that both workers and firms bargain for their share of the surplus (see wage function lectures), the wages are determined by Nash bargaining. Additionaly, other condition should be met: $\beta = \frac{u}{m} \cdot \frac{\partial m}{\partial u} = \eta_u$, where β is the workers' bargaining power and η_u is the elasticity of the matching function w.r.t. unemployment

4) Recall $\theta = \frac{V}{U}$, $\varepsilon = -\frac{d\ln V}{d\ln U} = -\frac{dV}{V}\cdot\frac{U}{dU} = -\frac{dV}{dU}\cdot\frac{1}{\frac{V}{U}} = -\frac{V'(u)}{\theta}$ $\therefore$ $\varepsilon = \frac{-V'(u)}{\theta}$

$$sw(u) = 1 - (1-z)\cdot u - V(u)\cdot r$$

$$\frac{dsw}{du} = -(1-z) - V'(u)\cdot r = 0$$

$$V'(u) = -\frac{(1-z)}{r}$$

$$\downarrow$$

$$\theta \cdot \frac{V'(u)}{\theta} = -\frac{(1-z)}{r}$$

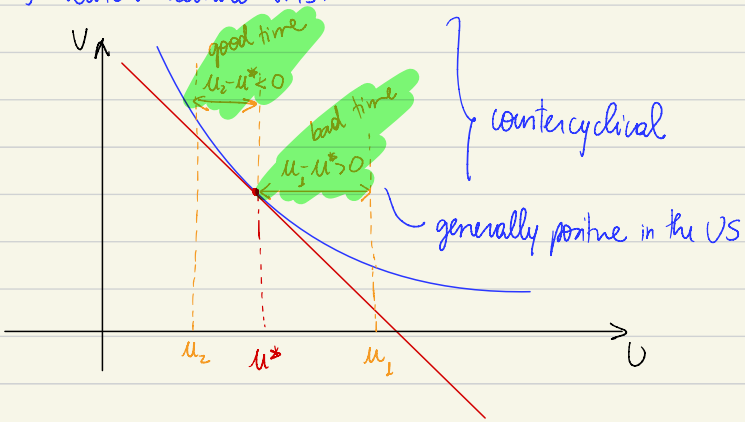
$$\theta \cdot \varepsilon = \frac{1-z}{r}$$

$$\theta = \frac{1-z}{\varepsilon \cdot r}$$

Now, using the notation adopted in this question:

$$\theta = \frac{1-G}{\varepsilon \cdot K}$$

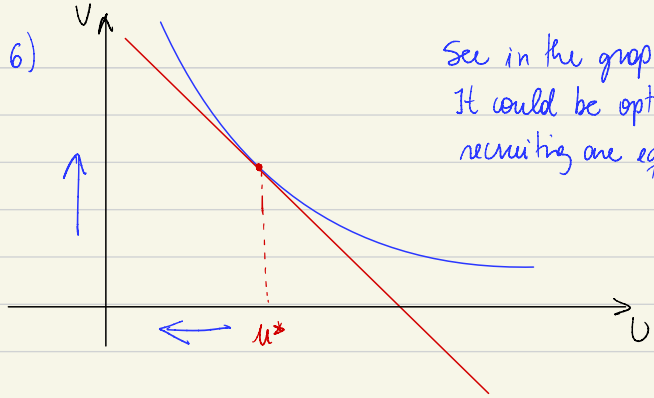
5) watch lecture V.93.



countercyclical

generally positive in the US

6)



See in the graph: as you decrease u, V goes up. It could be optimal if the costs associated with recruiting are equal to zero. See question #1.

7) Hosios: $\beta = \eta_U$ or $1-\beta = \eta_V$

8) $\theta = \dfrac{1-z}{\varepsilon \cdot r}$

$\uparrow z \rightarrow \downarrow \theta$

$\uparrow r \rightarrow \downarrow \theta$

$\uparrow \varepsilon \rightarrow \downarrow \theta$